
Temperature & Moisture Advanced Technology Evolution (TMATE)

Other Transaction Authority (OTA)

Project Competitive Announcement

**Request for Project Proposal (RPP)
Supporting Research & Development Efforts
for
Office of Low Earth Orbit (LEO) Observations**

RPP No.: 2026-NESDIS-BB-0001

Title:

Temperature & Moisture Advanced Technology Evolution (TMATE)

Issued by:

National Oceanic and Atmospheric Administration (NOAA)
Acquisition and Grants Office (AGO)
Satellite Information and Acquisition Division (SIAD)
SSMC 2, BLDG 02
1325 East-West Highway, Silver Spring, MD 20910

For the
Office of Low Earth Orbit (LEO) Observations
National Environmental Satellite, Data, and Information Services (NESDIS)
8800 Greenbelt Rd,
Greenbelt, MD 20770

PROPOSAL RESPONSE DATE: MARCH 16, 2026 at 3:00PM ET/12:00PM PT

TABLE OF CONTENTS

INTRODUCTION	3
OBJECTIVES	5
HOW CAN YOU PARTICIPATE?	5
PROCESS AND BASIS FOR EVALUATION	6
Phase I: Request for White Papers (RWP)	6
Phase II: Final Down-Select	9
APPENDIX 001: STATEMENT OF OBJECTIVES	11
APPENDIX 002: PROPOSAL REQUIREMENTS AND DISCLAIMERS	12
APPENDIX 003: PHASE SUBMISSION GUIDELINES	14
Phase I: White Paper Format	14
Phase II: Final Down-Select Format	16

INTRODUCTION

The National Environmental Satellite, Data, and Information Services (NESDIS) within the National Oceanic and Atmospheric Administration (NOAA), seeks to enter into one or more Other Transaction Agreements (OTAs) pursuant to the authority granted under Section 301 of the Weather Research and Forecasting Innovation Act of 2017 (15 U.S.C. 8531) as amended by the National Integrated Drought Information System Reauthorization Act of 2018, here-to-forward referred to as “the Act”, as amended by Section 503 of the National Oceanic and Atmospheric Administration Commissioned Officer Corps Amendments Act of 2020 (Pub. L. No. 116-259, title V, § 503, 134 Stat. 1153 (2020)). To stimulate and advance microwave sounding capabilities, NOAA is interested in pursuing existing industry-led microwave-based concepts, designs, and/or on-orbit assets to identify and evaluate temperature and moisture radiance profile collection solutions in polar orbit with operational readiness by 2030. Under this OTA, NOAA will support formulation activities, technology maturation, and architecture/design maturation of microwave sounding data sources.

Under Section 301 (d) of the Act, “In order to enhance the effectiveness of data and satellite systems used by NOAA to meet its missions, the Under Secretary may enter into and perform such transaction agreements on such terms as the Under Secretary considers appropriate to carry out basic, applied, and advanced research projects to meet the objectives described in the Act. The transaction agreements authorized are limited to research and development activities and may only be used as follows:

- i. For the construction, use, operation, or procurement of new, improved, innovative, or value-adding satellites, instrumentation, ground stations, and data;
- ii. To make determinations on how best use existing or planned data, systems, and assets of NOAA; and,
- iii. Only when the objectives of NOAA cannot be met using a cooperative research and development agreement (CRADA), grant, procurement contract, or cooperative agreement.

NESDIS, located at NOAA headquarters in Silver Spring, MD, is responsible for collecting and providing the critical satellite Earth observations and other essential environmental information needed for disaster preparedness, all hazards response and recovery, and the protection of the Nation’s critical infrastructure and natural resources. The mission of NESDIS is to provide secure and timely access to global environmental data and information from satellites and other sources to promote and protect the Nation's security, environment, economy, and quality of life. The purpose of this OTA directly aligns with the mission of NESDIS. NESDIS’ Office of Low Earth Orbit Observations (LEO), Near Earth Observations Network (NEON) Program, which has the responsibility to provide the next generation of operational polar meteorological observation data via a constellation of disaggregated sensors and services following the completion of the Joint Polar Satellite System (JPSS) mission.

This announcement uses a multiple-phase competitive down-select that **may** result in the award of a single project or multiple projects. Dates listed in this announcement are tentative and subject to change based on actual progress made through each phase of

selection. This project is entered to solve the capability gap through a pilot or novel application of commercial technologies.

**PROJECT AWARD IS SUBJECT TO THE AVAILABILITY OF FUNDS AT THE
TIME OF AWARD**

OBJECTIVES

Problem Statement: Atmospheric microwave sounding data is a critical input to meteorological numerical weather prediction models.

Capability Gap: The Government is interested in exploring non-traditional sources of commercial data to augment Government data from flagship instruments from legacy and future systems, currently operating in the 1330 and 2130 Local Time of Ascending Node (LTAN). Data provided under this agreement will be delivered to NOAA and released to the general public.

HOW CAN YOU PARTICIPATE?

1. Review the Process and Basis for Evaluation section.
2. Review Appendix 001: Statement of Objectives.
3. Review Appendix 002: Proposal Requirements and Disclaimers.
4. Review Appendix 003: Phase Submission Guidelines.

PROCESS AND BASIS FOR EVALUATION

Phase I: Request for White Papers (RWP)

Due date: MARCH 16, 2026 at 3:00PM ET/12:00PM PT

The purpose of this phase is to discover, learn, and understand more about the proposed solution through the evaluation of White Papers. This phase is intended to explore the capability the proposed solution offers the Government. The Government encourages offerors to focus on the innovative approach the proposed solution brings for consideration by the Government, and other capabilities the proposed solution may offer beyond what the Government requires.

Offerors must submit White Papers by the date and time due in the format provided in this announcement. Submissions not conforming to the designated format or received after the due date and time will not be accepted.

The Government requires Rough Order of Magnitude (ROM) pricing during this Phase. The ROM should include a total price summary of the solution. It is anticipated any award under this agreement will be Firm Fixed Price (FFP).

Offerors shall refer to APPENDIX 003: PHASE SUBMISSION GUIDELINES for White Paper format.

During this phase, the Government may correspond with the offeror to exchange additional information, explanation, material, clarifications, or discussions of the solution.

The Government may issue individual offerors additional instructions and requirements or provide or request supplementary information based on the Government's desire to further explore the solution for adoption to the Government's requirements.

The offeror may be required to provide a detailed pricing solution for further evaluation if requested by the Government.

White Papers shall be submitted by the due date and time to brooke.bernold@noaa.gov and anthony.eubanks@noaa.gov for collection and evaluation.

Phase I - Basis for Evaluation:

White Papers shall be evaluated using selections from the following criteria. Solutions may be qualitatively or quantitatively assessed by Government evaluators. Evaluation results are based on evaluator interpretations of information submitted. The Government's interpretation of offered solutions determines favorability and

acceptability. Solutions will be evaluated using selections from the following criteria:

A. The Government will assess if the proposed solution is capable of meeting objectives identified in Appendix 001.

Favorable: Existing and adaptable commercial or other solution that meets or exceeds all objectives identified in Appendix 001 without what the Government interprets to be significant development.

Potential: Existing and adaptable commercial or other solution that meets all objectives identified in Appendix 001; however, requires what the Government interprets to be significant development.

Unfavorable: Existing and adaptable commercial solution that does not meet all objectives identified in Appendix 001 and requires what the Government interprets to be significant development.

B. Rough Order of Magnitude (ROM) Pricing

Favorable: Solution ROM pricing meets Government budget constraints and does not have excessive ongoing license or maintenance cost.

Potential: Solution ROM pricing meets Government budget constraints.

Unfavorable: Solution ROM pricing may meet Government budget constraints, but costs are unclear or excessive, or the solution ROM does not meet Government budget constraints.

Note: Government budget constraints will not be provided, the offeror must propose a realistic ROM based on the offered solution.

Phase I - White Paper Down-Selection:

- A. Offerors with a White Paper rated “Favorable” in all criteria will be placed into **Category 1 – Selected Offerors** and receive an invitation for advancement to Phase II. Offerors with White Paper ratings of “potential” or “unfavorable” **may** be placed in this category at Government discretion should the Government desire to further explore the solution.
- B. Offerors not favorably evaluated will be placed in **Category 2 – Non-Selected Offerors** and removed from competition.
- C. The Government may provide feedback to Category 1 and Category 2 offerors at its discretion.

Phase II: Final Down-Select

Due Date: **To be provided with the invitation to Phase II**

Awards under this announcement will be OTAs under the authority granted under Section 301 of the Weather Research and Forecasting Innovation Act of 2017 (15 U.S.C. 8531) as amended by the National Integrated Drought Information System Reauthorization Act of 2018, here-to-forward referred to as “the Act”, as amended by Section 503 of the National Oceanic and Atmospheric Administration Commissioned Officer Corps Amendments Act of 2020 (Pub. L. No. 116-259, title V, § 503, 134 Stat. 1153 (2020)). Multiple awards may be made from this announcement should it be in the Government’s interest and funds are available.

All potential offerors shall be aware that funding in any or all areas may change with little or no notice, impacting the number of award(s).

The Government reserves the right to select all, part, or none of the offered solutions. Success Criteria for individual project(s) will be determined in Phase II.

The Government may allow offerors to collaboratively build the Agreement for their specific solutions with the Government team.

Phase II - Basis for Evaluation

- A. The offeror(s) selected for invitation to Phase II must be registered in [SAM.gov](https://sam.gov) to be eligible for agreement award.
- B. All solutions, as detailed in the completed Agreement, may be qualitatively or quantitatively assessed by Government evaluators. Evaluation results are based on evaluator interpretations of information presented. The Government’s interpretation of offered solutions determines favorability and acceptability. Offerors will be evaluated using selections from the following criteria for final project agreement award. Evaluation criteria are not in any order of priority.
 - 1) **Price**
 - 2) **Schedule**
 - 3) **Data Rights/Intellectual Property**
 - 4) **Applicability of the solution to the Statement of Objectives in Appendix 001**

Acceptable: The final negotiated solution is technically acceptable to the Government and pricing is determined by the Government to be fair and reasonable.

Unacceptable: The Government is unable or unwilling to accept the final technical solution as technically acceptable, or pricing is not determined to be fair and reasonable by the Government.

Phase II Selection:

- A. Offerors with an evaluated solution as “Acceptable” will be placed into **Category 1 – Selected Offerors** and receive an awarded Agreement.
- B. Offerors receiving an “Unacceptable” rating will be placed in **Category 2 – Non-Selected Offerors** and removed from competition.
- C. The Government may provide feedback to Category 2 offerors at its discretion.

APPENDIX 001: STATEMENT OF OBJECTIVES

Provided as a separate attachment. See Attachment: Appendix 001: Statement Of Objectives

APPENDIX 002: PROPOSAL REQUIREMENTS AND DISCLAIMERS

- A. Competitive procedures will be used for this acquisition to the maximum extent practicable. This announcement serves as a competitive opportunity for interested parties to present solutions for evaluation and potential selection for a project and is considered to satisfy the reasonable effort to obtain competition.
- B. The Government will not reimburse interested offerors for costs to prepare and submit any aspect of proposal, pricing information, travel, or any other activity during the competitive selection process.
- C. Unnecessarily elaborate brochures or marketing materials are not desired.
- D. Use of diagrams or figures to depict the essence of the proposed solution is encouraged.
- E. All information in the proposed solution must be unclassified.
- F. The delivery period for any project awarded as a result of this announcement shall be negotiated; however, it is not intended to exceed 24 months.
- G. Proposals from offerors in Category 2 may be considered by the Government for a potential solution up to one (1) year after submission for the same or similar requirements.
- H. Offerors shall not submit proposals on behalf of another entity. The offering entity must be eligible for award and provide the solution. Entities requiring a reseller or distributor that is not identified at the time of initial proposal as the primary offeror will be removed from competition. For proposals with teaming or partnership arrangements, the primary offeror shall be clearly identified.

Offerors shall not directly contact the Requiring Activity or other Government activities related to this announcement. Offerors shall direct all communications through NOAA via the following points of contact:

Brooke Bernold	brooke.bernold@noaa.gov
Anthony Eubanks	anthony.eubanks@noaa.gov

ALL QUESTIONS REGARDING THIS WHITE PAPER REQUEST MUST BE SUBMITTED NO LATER THAN 27 FEBRUARY 2026 AT 12:00 EST. NO QUESTIONS WILL BE CONSIDERED AFTER THIS DATE. ANSWERS TO THE QUESTIONS WILL BE PROVIDED AS AN AMENDMENT TO THIS ANNOUNCEMENT.

- I. **Project Awards.** Project award(s) as a result of this announcement will be in the form of Project Agreement (PA) between NOAA and the selected offeror. Flow-down articles may be provided at the offeror's request. OTA Articles to be included in any

Project Award document (PA) between NOAA and the selected offeror are not subject to revision, modification, or deletion.

- J. **Feedback.** Participants in OTA selections are not provided debriefs. The Government **may** provide feedback at its discretion to selected and non-selected offerors. The Government **may** provide solution feedback at its discretion to awardees that successfully or unsuccessfully complete projects. Feedback is a Government option and not required.
- K. **Use of Contractor Support.** The Government may utilize the Contractors and/or Consultants listed below to assist in the evaluation of the proposals for this OTA as non-voting subject matter experts. These Contractors/Consultants will have access to any and all information submitted by Offerors, and will be subject to the restrictions contained in the following clauses: CAR 1352.209-71 – Limitation of Future Contracting; CAR 1352.209-72 – Restrictions Against Disclosure; CAR 1352.209-74 – Organizational Conflict of Interest; and FAR Clause 52.203-16 – Preventing Personal Conflicts of Interest. The companies/organizations employing the subject Contractors/Consultants are also required to comply with these same confidentiality restrictions. All federal and non-federal personnel participating in the evaluation of this proposal will have executed Non-Disclosure Agreements.
1. Aerospace Corporation
2310 E. El Segundo Blvd,
El Segundo, CA 90245
 2. Ascendant Engineering, LLC
1455 Coventry Meadows Drive
Sykesville, MD 21784
 3. Tecolote Research Incorporated
420 S. Fairview Avenue, Suite 201
Goleta, CA 93117-3654
 4. Science and Technology Corporation (STC)
21 Enterprise Parkway, Suite 150
Hampton, VA 23666

APPENDIX 003: PHASE SUBMISSION GUIDELINES

Phase I: White Paper Format

White Papers shall adhere to the following required format:

1. **Specifications:** White Papers shall conform to the following specifications.
 - a. Font size shall be no smaller than 10-point with one-inch margins, and one inch header and footer.
 - b. Length is limited to **TEN PAGES MAXIMUM**. The Cover Page is not included in the page count.
 - c. The White Paper shall be submitted in Adobe Portable Document Format (PDF) and native format (e.g., Microsoft Word, Excel).
2. **Cover Page:** Offerors shall provide the following background information with their white paper submissions.
 - Offeror's Company Name
 - Company Address
 - Company Point(s) of Contact (name, email, phone)
 - Where is your organization based? (US or Internationally Based)
 - Business size: Large or Small Business, or not a commercial endeavor
 - System for Award Management Unique Entity Identifier (SAM UEI)
 - Commercial and Government Entity (CAGE) Code
3. **Executive Summary:** A brief description of your organization and core competencies.
4. **White Paper Solution:** Describe the unique aspects of your solution as it relates to the objectives in Appendix 001.
5. **Value Proposition Summary:** It is the offeror's responsibility to demonstrate why it is offering the best solution and what value-add this solution will bring to NOAA. This is the offeror's chance to convince the Government as to why it should invest in a solution. Offerors are reminded this is not a sales pitch and should focus on demonstrating why the proposed solution will address the objectives. The proposition should be written in clear, concise, common language statements. ROM pricing must be included in the White Paper.
6. **ROM:** Provide Rough Order of Magnitude (ROM) pricing during this Phase. The ROM should include a total price summary of the solution. It is anticipated any award under this agreement will be Firm Fixed Price (FFP).

7. Offeror Contribution: Describe the Offeror's proposed contribution to the agreement. Contributions may include cash contributions or in kind contributions.

"Cash Contribution" means the Prime's financial resources expended to perform this Agreement. The cash contribution can be derived from Prime's funds or outside sources or may also come from non federal contract or grant revenues or from profit or fee on a federal procurement contract. A Prime's own source of funds may include corporate retained earnings, current or prospective Independent Research and Development (IR&D) funds or any other indirect cost pool allocation. New or concurrent IR&D funds can be utilized as a cash contribution provided those funds identified by the Prime are to be spent on the conduct of the Agreement. Cash contributions include the funds a Prime will spend for labor (including benefits and direct overhead), materials, new equipment (prorated if appropriate), and sub to the prime efforts expended on this Agreement.

"In Kind Contribution" means the Prime's nonfinancial resources expended by the Prime, such as wear and tear on in-place capital assets like machinery or the prorated value of space used for the conduct of the activity, and the reasonable fair market value (appropriately prorated) of equipment, materials, and other property used in the conduct of the activity.

Phase II: Final Down-Select Format

Offerors selected for invitation to Phase II shall provide a complete solution, detailed pricing, and other supporting information as required by the Government. Offerors will be issued a Project Agreement template and shall incorporate their solution into a final Project Agreement format. The Government intends to issue the Project Agreement template as a Request for Proposal (RFP); however, **may** elect to collaboratively develop the Project Agreement with the selected offeror(s).

The offeror shall propose the total price to complete the project agreement, broken out to show the Government-funded price portion and the Company-funded contribution, including the value of non-labor contributions to the completion of the project. The Company shall provide any other data or supporting information as the parties agree is necessary for the determination of a fair and reasonable price. A milestone schedule shall be included which clearly indicates the completion of priced tasks associated with the milestone payment schedule. For firm fixed price agreements, the milestone schedule will serve as a payment schedule for any subsequent award.

- a) Full technical response. Describe the solution, capabilities, critical cost, and schedule drivers. Length is limited to **TWENTY PAGES MAXIMUM**. The Cover Page is not included in the page count.
- b) Price and Schedule. Provide price(s), total price, and a sequential milestone payment and delivery schedule with technical milestones outlined.
- c) Data and Intellectual Property Rights. Clearly describe all intellectual property and technical data to be delivered and associated rights. Copies of license agreements shall be provided.
- d) Award Eligibility. Each respondent shall provide its business status and its eligibility to receive an award, registration in SAM.gov.
- e) Entity Identification. Company Name, Commercial and Government Entity (CAGE) code, System for Award Management Unique Entity Identifier (SAM UEI), address, Point(s) of contact, phone number, email address, and organizational charts shall be included for all significant participants.
- f) Document deliverables. Identify documents, to include a legend, which are being submitted as a deliverable. Proprietary documents must be clearly marked.